

Homework 06

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October 4, 2026

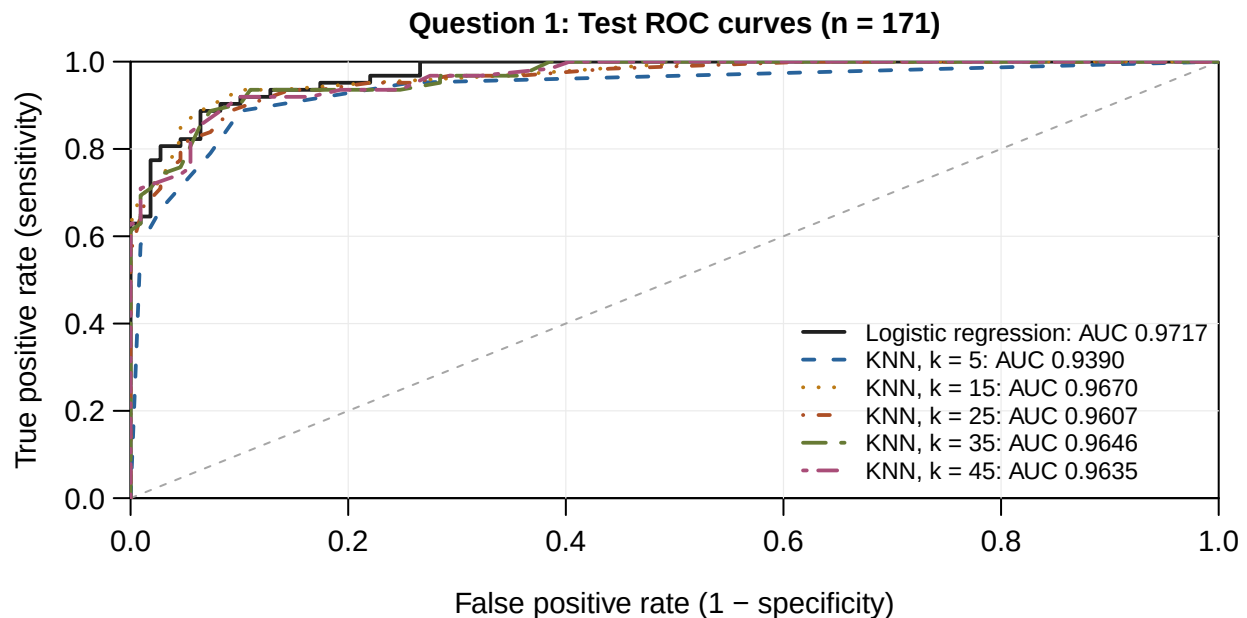
Question 1: Compare KNN and logistic regression

Dataset	Total	Benign	Malignant
Training	398	248	150
Test	171	109	62

I used seed **432061** and randomly selected 398 of the 569 observations for training; the remaining 171 were used for testing. All models in both questions use this same split. The three predictors were standardized using only the training means and sample standard deviations. The positive class is malignant ($y = 1$).

Model	Test AUC
Logistic regression	0.9717
KNN, $k = 5$	0.9390
KNN, $k = 15$	0.9670
KNN, $k = 25$	0.9607
KNN, $k = 35$	0.9646
KNN, $k = 45$	0.9635

Logistic regression includes an intercept and has no penalty. Each KNN probability is the fraction of malignant cases among the selected neighbors. ROC curves vary the probability cutoff; higher scores always indicate the positive class. AUC uses trapezoidal integration and is checked against the rank formula, with half credit for tied scores.

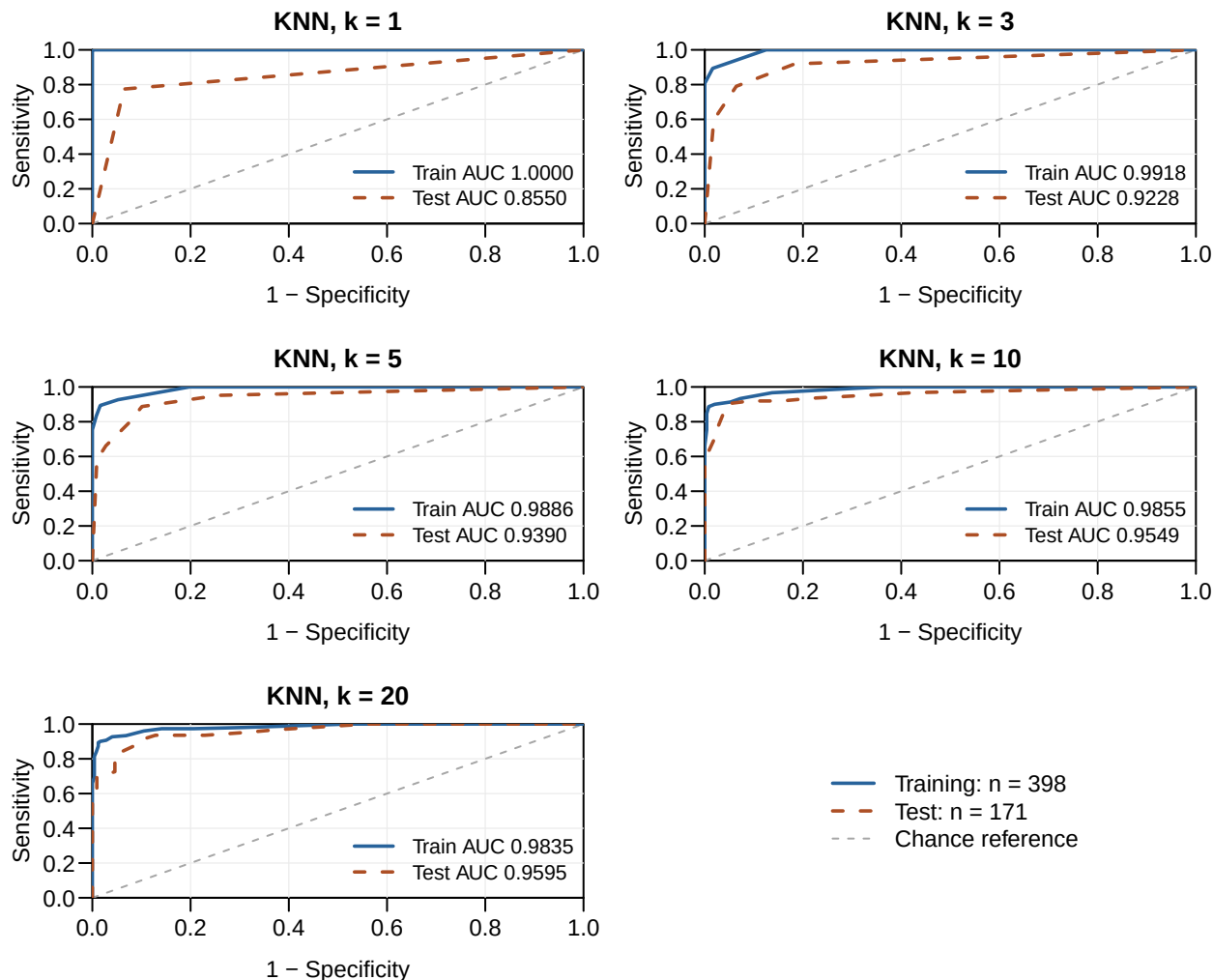


Model preference and justification: In this division, I chose logistic regression because its test AUC was 0.9717, which was the highest. Among KNN, the best value was $k=15$, with an AUC of 0.9670; the difference between the two was approximately 0.0047. This indicates that the sorting performance of logistic regression in distinguishing between malignant and benign cases is still acceptable.

Question 2: Training and test ROC curves

The training and test sets, standardization, distance, and weights are the same as in Question 1. For training predictions, each observation is included among its own nearest neighbors.

k	Training AUC	Test AUC	AUC gap
1	1.0000	0.8550	0.1450
3	0.9918	0.9228	0.0689
5	0.9886	0.9390	0.0497
10	0.9855	0.9549	0.0306
20	0.9835	0.9595	0.0240



Training versus test performance, the effect of k , and $k = 1$: I found that in this experiment, the training AUC for each k was higher than the test AUC. Especially when $k = 1$, the nearest neighbor for each training point is itself, so the predicted probability is the same as the original label, and the training AUC reached 1. However, the test AUC was only about 0.8550, indicating that its good performance on the training set does not represent the actual prediction performance. Only looking at the training AUC would overestimate the performance of this model. When k increased from 1 to 20, the training AUC dropped from 1.0000 to 0.9835, and the test AUC rose from 0.8550 to 0.9595. The gap between the two became smaller. With more neighbors considered, the prediction results are less affected by a single point, and the test performance is also better. Therefore, among the several k values compared in this experiment, I would choose $k = 20$, because its test AUC is the highest and the gap between the training and test results is the smallest.

Question 3: Peer evaluation of Homework 5

Assignment: Homework 6 peer grading assignment.

Assigned reports: krishay-b and mazy2005.

1. Identify the mistake or item for improvement

Selected report and question: krishay-b's Homework 05, Question 2(c).

Specific mistake or item for improvement: The explanation claims that correlation in Setting 2 lets a linear lasso model capture some of the signal from the quadratic term X_3^2 . This claim should be corrected.

2. Explain your assessment

In Setting 2, correlation also lets lasso leak some of the X_3^2 signal into the shared factor, but only as a linear approximation.

you couldn't say it under this question's condition, since the occasional correlation in this sample couldn't be the proof of this mechanism.

3. Provide a correction

In Setting 2, the correlation among the predictors does not eliminate the limitations of the lasso model. Because the model includes only linear terms, it cannot capture the nonlinear effect of X_3^2 . Therefore, we cannot say that lasso has captured the signal from the quadratic term.

Posted GitHub Issue: Peer review of krishay-b's Homework 05, Question 2(c).